

Frontier *(working title)*

An all-in-one research assistant for builders — one knowledge substrate that levels up both you and your agents.

Track: Product · **Team:** 3 people · **Demo Day:** September 9, 2026

The Problem

Builders need to get to the frontier of a topic fast — the science, the evidence, the techniques, the market — and today that means a dozen tabs: Google Scholar, arXiv PDFs read one by one, competitor pricing pages, Reddit threads. The friction isn't finding information; it's the constant context switching and the slow, manual work of synthesizing scattered sources into working expertise.

The problem has doubled. Builders now need to level up **two** learners: themselves, and their AI agents. The research you do to understand a domain and the context your agents need to perform well in that domain are the same knowledge — but no tool treats them that way. Elicit and SciSpace help humans read papers; retrieval pipelines feed agents raw chunks. Nothing bridges them.

The Thesis: Context Is the Product

There are two kinds of context, and one system should produce both:

- **Human context** — helping a person find and learn information as fast as possible, with minimal friction and context switching. This spans *academic/research context* (papers, evidence, concepts, libraries, architectural approaches) and *market context* (markets, competitors, pricing).
- **Agentic context** — finding and distilling the best research into the highest-quality context for an agent working in a particular domain or project.

Our core design decision: **one living knowledge base per project, two views**. You point the system at a topic; it ingests and synthesizes sources into a structured, citation-grounded knowledge base. A human view makes learning it fast and intuitive. An agent view exports the same knowledge as high-quality agent context. No existing product cleanly solves both.

What We're Building

The substrate: a living, cited knowledge base

Ingestion from academic APIs (Semantic Scholar, arXiv), user-uploaded PDFs, and live web sources. An LLM synthesis pipeline distills these into interlinked topic pages where **every claim links to the exact passage in its source**. The citation graph ranks what's foundational versus what's frontier.

The human view: map and terrain

- **Map — synthesis pages:** per-topic digests of the state of the art, key results, open questions, and disagreements between papers — so you navigate the frontier instead of reading papers one by one.
- **Terrain — the augmented reader:** our signature interaction. Read any paper in-app; select any term or claim and get an inline explanation, definition, or "what do other papers say" — powered by the knowledge base, with zero tab-switching.
- **Path to the frontier:** a generated reading path per topic, ordered from foundational to frontier using the citation graph and concept dependencies.
- **Market map:** the same pipeline pointed at the market — a synthesized, source-cited page of competitors, pricing, positioning, and gaps, browsable in the same interface.

The agent view: distilled packs + live retrieval

- **Context packs:** a curated, distilled set of markdown files — domain overview, key concepts, best practices, pitfalls, market landscape, all with citations — dropped straight into an agent's context (e.g., as a CLAUDE.md/skill for Claude Code).
- **MCP server:** the live drill-down layer — the agent queries the knowledge base mid-task for specific claims with citations, mirroring the human map-and-terrain design.

Why It's Hard to Copy

A weekend clone is a chat wrapper over search — that's not this. The defensible core is the **single substrate serving both consumers**: a synthesis pipeline whose output is simultaneously (a) trustworthy enough for a human to learn from, because every claim is one click from its source passage, and (b) structured enough to measurably improve agent performance. The citation-grounded synthesis, the citation-graph-derived frontier paths, and the round trip from human-verified knowledge to agent context are real engineering with compounding quality — not a prompt.

Validation Plan

Claim	Test	Measure
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Context packs level up agents	A/B benchmark: same agent (Claude Code), same builder tasks in a chosen domain, with vs. without the pack	Rubric-scored output quality: correctness, use of domain best practices, citation accuracy
Humans reach the frontier faster with less friction	Timed within-subjects study, ~6 recruited classmates: answer frontier questions on a topic using our tool vs. their normal workflow	Time to answer, answer accuracy, count of context switches (tab switches)
Synthesis is trustworthy	Citation-faithfulness spot checks: sampled claims verified against their cited passages	% of sampled claims supported by their source

User feedback from the study is the validation evidence presented at Demo Day, per the product-track requirements.

12-Day Plan

Days	Goal
1–4	Substrate. Academic ingestion (Semantic Scholar/arXiv + PDF upload through one paper pipeline), synthesis into the knowledge base, first cited map pages.
5–8	The two views. Augmented reader with inline lookup; path-to-frontier; market-map adapter and page; context-pack export + MCP server; deployed.
9–12	Proof. Feature freeze → run the user study and agent A/B benchmark, fix what testing exposes, build the deck.

Team split: one on the ingestion/synthesis pipeline (academic + market adapters), one on the web app (map, reader, paths), one on agent exports (packs, MCP) and evals.

Stretch goals: technical-web ingestion (library docs, GitHub, engineering blogs); adaptive explanation depth calibrated to the reader's level (an area we want to explore/research).

Build-in-Public Plan

Each of the 3 team members posts on X throughout the build (5+ posts each), sequenced to the plan: (1) the thesis — two kinds of context, one substrate; (2) first synthesized frontier map from real papers; (3) the augmented reader inline-lookup clip; (4) the agent A/B result — same agent, with and without the pack; (5) demo-day launch. Targets per person: 150+ engagements, 25+ followers, ≥1 external repost.

The Demo

One loop, end to end, in under ten minutes: pick a topic → watch the frontier map and market map build with citations → read a paper with inline concept lookup, no tabs → generate the path to the frontier → export the context pack → run the same agent task with and without it and show the difference — then the numbers from the user study.